PAPER_722: UQFF Quantum Variable Documents: M_bh, mu_j, P_core, t_n, pi

Class: `UQFFKnowledgeBaseKB8` **CP4 Entry:** #306 **Keywords:** UQFF, quantum variables, black hole mass, magnetic moment, P_core, normalized time, pi quantum encoding, SMBH **Session:** 176 | **Version:** v5.33 **Source:** grok_share_ba508f76c8e.txt

Abstract

Five quantum variable document pages define: M_{bh} (Milky Way SMBH mass for U_g4 scaling), μ_j (magnetic string dipole moment), P_{core} (core pressure normalization for U_g3), t_n (normalized phase time modulating all UQFF terms via $\cos(\pi t_n)$), and π (the mathematical constant encoding 26 quantum states through $[SSq]^{n_{26}} e^{-\pi - t}$).

1. Variable Registry

Variable	Value	Physical Role
M_{bh}	1.989×10^{36} kg ($\sim 10^6 M_\odot$)	Scales U_g4 AGN/Final Parsec
μ_j	3.38×10^{23} J·m	Magnetic string coupling in U_m
P_{core}	1.0 (normalized)	Core pressure factor in U_g3
t_n	0.0 s	Phase modulator: $\cos(\pi t_n)$
π	3.14159...	$[SSq]^{n_{26}} \cdot e^{-\pi - t}$, 26-state encoder

2. U_g4 Black Hole Scaling

$$U_{g4}(t) = k_4 \cdot \frac{\rho_{SCm} \cdot M_{bh}}{d_g} \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + f_{feedback})$$

3. U_g3 Star Formation

$$U_{g3} = k_3 \sum_j B_j(r, \theta, t, \rho_{SCm}) \cdot \cos(\omega_s t \pi) \cdot P_{core} \cdot E_{react}$$

4. Pi as Quantum Encoder

$$[SSq]^{n_{26}} \cdot e^{-\pi - t} \quad \text{encodes 26 quantum states via } \pi\text{-barrier}$$

References

- UQFF KB grok_share_ba508f76c8e.txt entry #72
- Session 176, v5.33

Whitepaper auto-generated by _gen_whitepapers_716_730.py – Star-Magic Session 176